

WHAT TO BUY, AND WHAT IT COSTS

AR-750

Everything for the AR-750, with a link to each shop.

There are two columns on the next pages. CHEAP is the least money that still works safely. BETTER is what I would spend if the money is there, and every line says why. You can mix them - most people buy cheap everywhere except the camera.

THE CHEAP BUILD

The parts on the list	Rs 18,560
Wire, screws, chassis, frame	Rs 4,420

TOTAL **Rs 22,980**

Works. Nothing in it is unsafe.

THE BETTER BUILD

The parts on the list	Rs 29,947
Wire, screws, chassis, frame	Rs 7,620

TOTAL **Rs 37,567**

Lasts longer and hurts less later.

THREE THINGS THAT CHANGE WHAT YOU BUY

The cheap 30 A brushed ESC is the wrong part.

The one everyone links to is a 4-8 V micro ESC for toy cars. At 12 V it dies on the first switch-on. Every hobby brushed ESC sold here stops at 3S, which is 12.6 V - under a freshly charged battery. So this list uses BTS7960 driver boards instead: 6-27 V, 43 A, Rs 256 each against Rs 7,685 for the one proper ESC that does the voltage. That saves about Rs 7,000, and I change the drive code to suit - say the word.

Farm spray nozzles are far too big for this rover.

It creeps down a row at 0.57 km/h. The smallest normal flat-fan nozzle would put down eight to twenty times too much. What actually fits is a 4 LPH drip emitter - 67 ml a minute - and fifty of them cost Rs 139. With the valves pulsing they sit at 39% duty, which is a clean band.

The Rs 162 pump will not open your valves.

It makes about 0.3 bar. The solenoid valves need 0.2 bar just to open and the nozzles want more. Spend the extra Rs 240 on the 100 psi pump or the boom will dribble.

BUY IT IN THIS ORDER, NOT ALL AT ONCE

- 1 Pi, SD card, buck converter** **about Rs 7,400**
Get the website running on your desk first. No rover needed.
- 2 Battery, switch, fuses, 2 drivers, 2 motors, wheels** **about Rs 6,700**
Now it drives. Put it on blocks and run bench.py wheels.
- 3 Servo, camera, cable, ADC, sensors** **about Rs 2,600**
Now it steers and sees.
- 4 Pump, valves, drippers, tank, tube** **about Rs 1,500**
The boom last - test it with plain water.

THE LIST

AR-750

Click any shop name to open the page.

PART	QTY	CHEAP	BETTER
THE BRAIN			
Raspberry Pi 4 Model B Runs the rover and serves the website. 2 GB is enough for what the software does today. Robu - 2 GB Robu - 4 GB - only worth it when you train the plant model	x1	Rs 6,449	Rs 11,469
MicroSD card 32 GB, class 10 A1 The Pi's disk. Cards are dear right now - this shop is half the rest of the market. ElectronicsComp - 655 + GST	x1	Rs 773	Rs 773
DRIVING			
BTS7960 43 A motor driver Drives one wheel motor, both directions. 6-27 V, 43 A - far more than these motors need. Robokits - cheapest of three shops Robu - same board, faster delivery	x2	Rs 512	Rs 618
12 V geared motor, 100 rpm, 27 kg.cm One per axle. This is the torque class that actually moves a loaded rover - the 300 rpm yellow motors are not. Robokits - RKI-1419	x2	Rs 1,270	Rs 1,270
Steering servo Points the front wheels through the linkage. Robocraze - MG996R - real torque is 11 kg.cm, tight but workable Robu - DS3218, 20 kg.cm - proper margin	x1	Rs 298	Rs 1,469
Rubber wheel 100 x 40 mm, 6 mm bore 40 mm wide matters more than you think on soft soil. Robocraze	x4	Rs 424	Rs 424
Stopping the wheels slipping on the shaft The wheels clamp on a 6 mm shaft with one grub screw. Under 27 kg.cm that screw rounds the shaft off. do it yourself - drill 3 mm through hub and shaft, drop a split pin in - costs nothing Robu - proper anti-slip coupling if you would rather not drill	x2	Rs 60	Rs 1,478
POWER			
12 V 18 Ah sealed lead acid battery The cheapest sensible pack by a mile. Heavy, but on a ground rover the weight is ballast you half want. Flipkart - Exide, about 6 kg Robu - LiFePO4 18 Ah, 2.5 kg, lasts years longer	x1	Rs 2,466	Rs 4,829
Battery charger 12 V 3 A Matches the lead acid pack. Robu - lead acid only Robu - also charges LiFePO4 - buy this if you might switch later	x1	Rs 639	Rs 1,607

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PART	QTY	CHEAP	BETTER
POWER			
Buck converter 12 V to 5 V, 5 A Feeds the Pi and the steering servo. 5 A, not 3 - the servo alone can pull 3 A against its stops. Robu	x1	Rs 185	Rs 185
Battery disconnect switch, 100 A Kills everything. Put it where you can reach it while walking beside the rover. Flipkart	x1	Rs 366	Rs 366
Inline blade fuse holders, waterproof, 5 pcs One fused branch per system, so one fault is one system. Robu	x1	Rs 589	Rs 589
Blade fuses, assorted 5-30 A 5 A on the lights and valves, 10 A on the pump and probe, 30 A on the main lead. Robu - about 11-20 each	x8	Rs 120	Rs 120
THE SPRAY BOOM			
12 V diaphragm pump, 100 psi, 3.1 l/min Self priming, and enough pressure to actually open the valves. Roboman	x1	Rs 399	Rs 399
12 V solenoid valve, 1/2 inch, normally closed One per boom section, so it can spray just the left, the middle or the right. Robu - the 1/4 inch brass one is 651 for no reason	x3	Rs 732	Rs 732
4 LPH drip emitters, pack of 50 THE NOZZLES. 67 ml/min each - see the nozzle page. Farm spray nozzles are 8 to 20 times too big for this rover. Flipkart - you need 5, you get 50 IndiaMART - 0.3 mm brass mist nozzles, 5 x 40 - a real atomised spray	x1	Rs 139	Rs 200
5 litre HDPE tank The spray tank. HDPE copes with garden chemicals. IndiaMART	x1	Rs 35	Rs 35
PVC tube 6 mm inside, per metre Tank to pump to valves to nozzles. Robocraze	x3	Rs 105	Rs 105
Stainless hose clamps Every joint. Spray leaks find every one you skipped. IndiaMART	x8	Rs 40	Rs 40
SEEING AND SENSING			
Raspberry Pi Camera v2 Follows the row AND judges the plants. The one part you should not cheap out on. Robu	x1	Rs 1,698	Rs 1,698

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PART	QTY	CHEAP	BETTER
SEEING AND SENSING			
CSI ribbon cable, 1 metre, 15 pin The camera sits 254 mm up at the front. The stock 15 cm cable will not reach. Probots - 15 pin is for Pi 4. A Pi 5 needs the 22 pin one	x1	Rs 199	Rs 199
ADS1115 4 channel ADC The Pi cannot read an analogue sensor at all. This board reads four. Robu	x1	Rs 139	Rs 139
Capacitive soil moisture sensor Goes on the end of the probe. Capacitive, NOT the cheap resistive kind - those corrode away in weeks. Robu	x1	Rs 47	Rs 47
ACS712 current sensor, 30 A Sees a jammed wheel seconds before a driver gets hot enough to fail. Robu	x1	Rs 72	Rs 72
MPU-6050 tilt sensor Without it the tilt cut-out can never fire, and this machine is tall on a narrow track. Robokits	x1	Rs 96	Rs 96
HC-SR04 range sensor Stops it before it drives into somebody. Needs two resistors - see the circuit diagram. Robu	x1	Rs 69	Rs 69
THE SOIL PROBE			
BTS7960 43 A driver (third one) Runs the probe's lead screw both ways. Robokits - same board as the wheels - buy three together	x1	Rs 256	Rs 256
Micro limit switch, roller lever One at each end of the probe's travel. Buy four, break two. QuartzComponents - Robu wants 155 each for the industrial version	x4	Rs 128	Rs 128
SWITCHING AND LIGHTS			
D4184 MOSFET module The Pi's pin cannot switch a pump or a valve. One of these does it for each - pump, three valves, and the lights. Robu - a module avoids the IRLZ44N / IRFZ44N mix-up	x5	Rs 175	Rs 175
12 V LED light So you can see what it is doing at dusk, and so the camera has light. IndiaMART - plastic, no IP rating IndiaMART - 20 W, waterproof - it lives outdoors	x2	Rs 80	Rs 360

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NO LINK - BUY THESE LOCALLY

Wire, screws and the frame are cheaper and faster from a hardware shop than posted to you.

Silicone wire, 10 AWG red and black, 2 m each

Rs 400

Rs 400

Battery to switch to fuse to the drivers.

Hook-up wire, 22 AWG, assorted

Rs 250

Rs 250

Every signal wire on the diagram.

Connectors - XT60 pair, spade terminals, ferrules

Rs 350

Rs 350

Flyback diodes 1N5408, 10 pcs

Rs 60

Rs 60

Across the pump and every valve. Four are essential.

Resistors 1k, 2k, 100k, 20k

Rs 60

Rs 60

The two dividers on the diagram.

Chassis - aluminium extrusion or plywood, wheels' mounts, screws

Rs 2,500

Rs 5,000

The cutting templates you already have are for sheet parts.

Steering linkage - rod ends, threaded rod, bearings

Rs 800

Rs 1,500

EVERYTHING

Rs 22,980

Rs 37,567

Add shipping. Most of these shops are free over Rs 500-1000, but you are ordering from six of them.

WHERE THE MONEY GOES, AND WHERE TO SAVE IT

AR-750

Read this before you order anything.

THE SIX LINES THAT ARE MOST OF THE MONEY

Raspberry Pi 4 Model B		Rs 6,449
12 V 18 Ah sealed lead acid battery		Rs 2,466
Raspberry Pi Camera v2		Rs 1,698
12 V geared motor, 100 rpm, 27 kg.cm		Rs 1,270
MicroSD card 32 GB, class 10 A1		Rs 773
12 V solenoid valve, 1/2 inch, normally closed		Rs 732

Those six are Rs 13,388 of the Rs 18,560 parts bill - 72% of it.

WHAT NOT TO SKIMP ON

The camera. Rs 1,698 and it does two jobs - follows the row and looks at the plants. A cheap USB webcam will focus wrong and colour wrong, and every plant decision comes through it.

The motors. The Rs 165 and Rs 170 geared motors you see everywhere are the 1.2 kg.cm class. They move a 2 kg tabletop robot. The Rs 635 one is 27 kg.cm, which is what this needs.

The pump. Rs 399 instead of Rs 162. The cheap one cannot open your own valves.

Fuses and the disconnect switch. Rs 950 all in. This is a machine with a 18 Ah battery that can dump hundreds of amps into a pinched wire.

WHERE YOU CAN SAVE, HONESTLY

Pi 4 with 2 GB, not 4 GB. Saves Rs 5,020. The software runs fine in 2 GB. Only go to 4 GB when you start training the plant model.

Lead acid, not lithium. Saves Rs 2,363. It is 3.5 kg heavier, and on a ground rover that is ballast, not a problem. Change safety.battery in config.yaml to warn 11.8 and stop 11.0 for a 12 V lead acid pack.

MG996R servo, not DS3218. Saves Rs 1,171. 11 kg.cm is tight but the steering linkage is not heavy. If it buzzes or gives up at full lock, that is the one to upgrade.

Drill and pin the wheels yourself. Saves Rs 1,418. A 3 mm hole through the hub and the shaft with a split pin beats a grub screw.

Skip the range sensor, encoders and tilt sensor at first. They are only Rs 165 together, so honestly do not - but the rover runs without them and the dashboard says so.

WHAT THIS LIST DOES NOT COVER

The frame. The cutting templates you already have are for sheet parts. Plywood and angle iron from a local shop is the cheap way; 20x20 aluminium extrusion is the tidy way. Budget Rs 2,500 to Rs 5,000.

The steering linkage. Rod ends, threaded rod and two bearings. Rs 800 to Rs 1,500 from a local hardware or an RC shop.

Spray chemicals. Whatever you actually spray, and the label on the bottle is what counts - not anything in this software.

Your remote. You already have the 6 channel set, so it is not on the list. You will need an SBUS inverter or a PPM output - see the circuit diagram.

One software change. The drive code as written today sends a servo pulse, because it was written for ESCs. The BTS7960 boards on this list want a PWM and a direction pin instead. That is a small change to ar750/hw/drive.py and one line in config.yaml, and I will make it for you - just ask. If you would rather not change anything, the part that works with today's code unmodified is the Cytron MDD30 at Rs 7,685, and it replaces both BTS7960 boards.